Machine Learning Lectures

Exploratory Data Analysis

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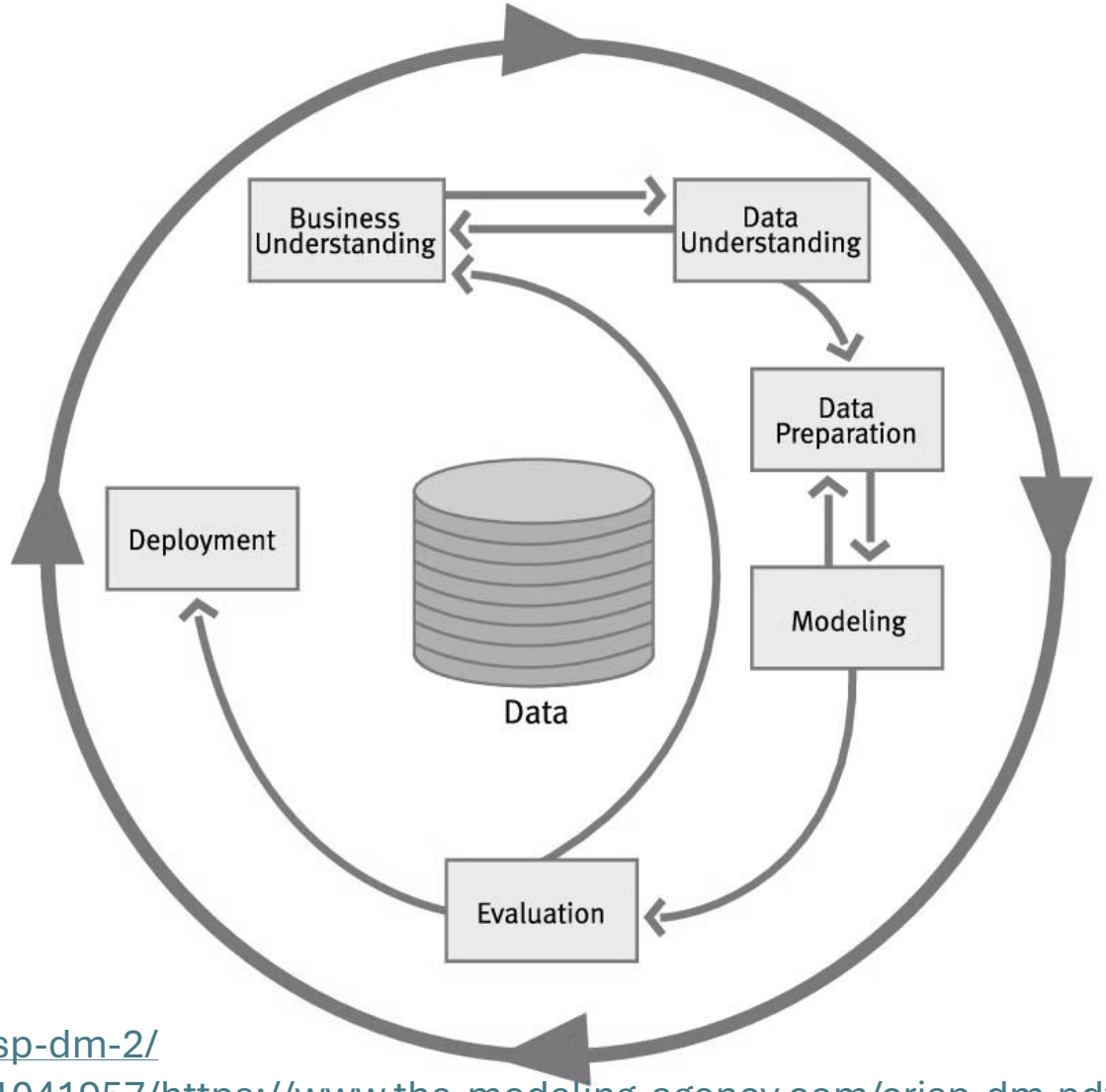
Agenda

- Introduction to EDA
- Basic Analytics
- Univariate, Bivariate, Multivariate Analysis
- Automation in EDA
- Visualization Tools and Resources
- Common EDA Challenges

Data Analytics pipeline: CRISP-DM



https://www.datascience-pm.com/crisp-dm-2/
https://web.archive.org/web/20220401041957/https://www.the-modeling-agency.com/crisp-dm.pdf

Business Understanding

Determine Business Objectives
Background
Business Objectives
Business Success Criteria

Assess Situation
Inventory of Resources
Requirements, Assumptions, and Constraints
Risks and Contingencies
Terminology
Costs and Benefits

Determine Data Mining Goals
Data Mining Goals
Data Mining Success Criteria

Produce Project Plan
Project Plan
Initial Assessment of Tools and Techniques

Data Understanding

Collect Initial Data
Initial Data Collection Report

Describe Data
Data Description Report

Explore Data
Data Exploration Report

Verify Data Quality
Data Quality Report

Data Preparation

Select Data
Rationale for Inclusion/ Exclusion

Clean Data
Data Cleaning Report

Construct Data
Derived Attributes
Generated Records

Integrate Data
Merged Data

Format Data
Reformatted Data

Dataset
Dataset Description

Modeling

Select Modeling Techniques
Modeling Technique
Modeling Assumptions

Generate Test Design
Test Design

Build Model
Parameter Settings
Models
Model Descriptions

Assess Model
Model Assessment
Revised Parameter Settings

Evaluation

Evaluate Results
Assessment of Data Mining Results w.r.t. Business Success Criteria
Approved Models

Review Process
Review of Process

Determine Next Steps
List of Possible Actions
Decision

Deployment

Plan Deployment
Deployment Plan

Plan Monitoring and Maintenance
Monitoring and Maintenance Plan

Produce Final Report
Final Report
Final Presentation

Review Project
Experience Documentation

What is EDA ?

Exploratory Data Analysis (EDA) is the process of **examining and analyzing datasets** to summarize their main characteristics, often with the help of **visualizations**.

EDA is typically performed before formal modeling or hypothesis testing, and its goal is to **uncover patterns**, **spot anomalies**, and **understand the distribution and relationships** between variables.

What is the purpose of EDA?

RAW DATA → → **CONCLUSIONS**

Exploratory Data Analysis (EDA)

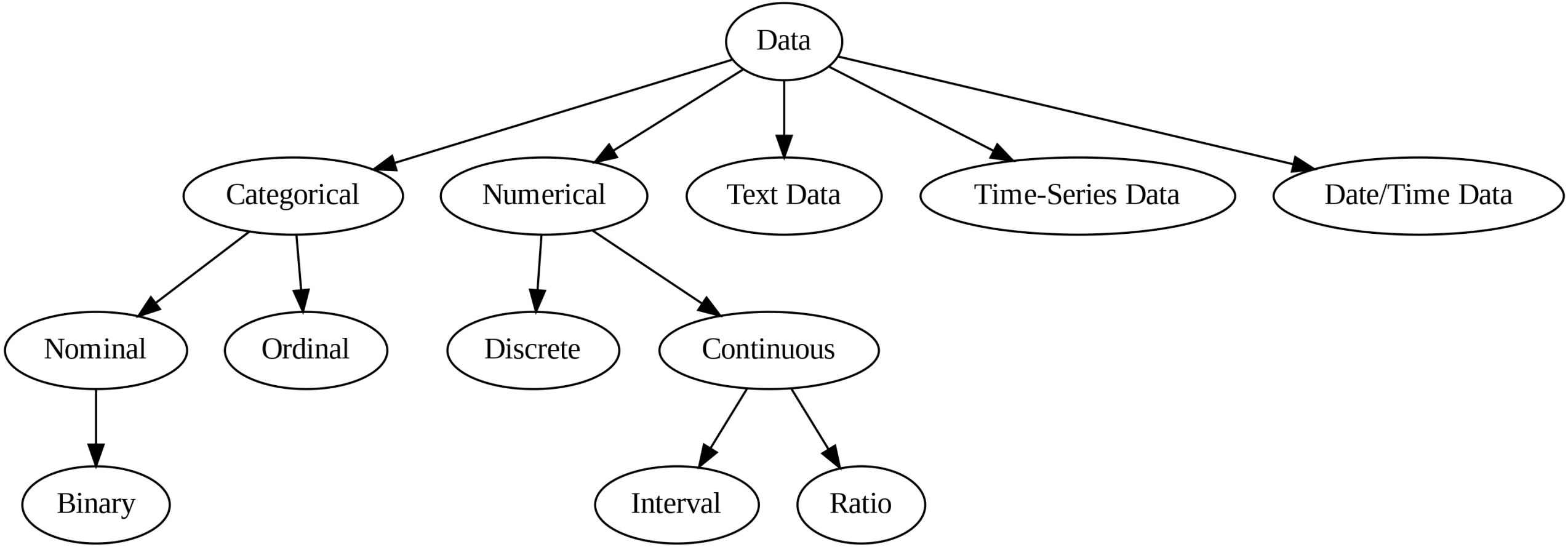
- Descriptive Analysis
- Adjustment of variable types
- Detection and treatment of missing data
- Identification of Atypical Data
- Correlation of Variables

Key points of EDA

Key Objective	Description
Understand Data Structure	Summarizes relationships between variables and visualizes the dataset structure.
Spot Patterns & Relationships	Identifies correlations, clusters, and trends.
Identify Outliers & Anomalies	Detects irregularities like outliers and missing data for proper handling.
Assess Data Distribution	Visualizes whether the data is normally distributed, skewed, or has multiple peaks.
Guide Feature Engineering	Helps in creating or transforming features to enhance model accuracy.
Handle Missing Data	Suggests ways to impute, remove, or clean missing values.
Reduce Errors	Minimizes the risk of incorrect assumptions by revealing hidden relationships.
Select the Right Model	Guides the choice of appropriate machine learning models based on the insights.

Data Types and Basic Analysis

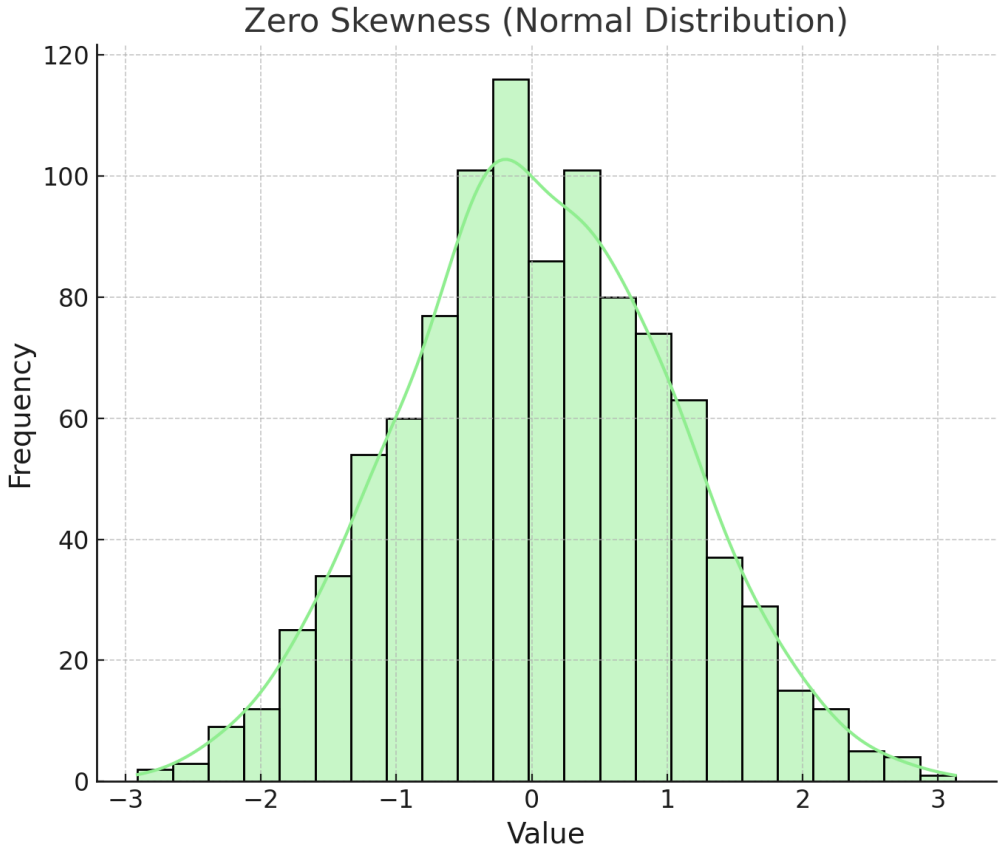
Most common types of data



Key Statistical Measures in EDA (numerical data)

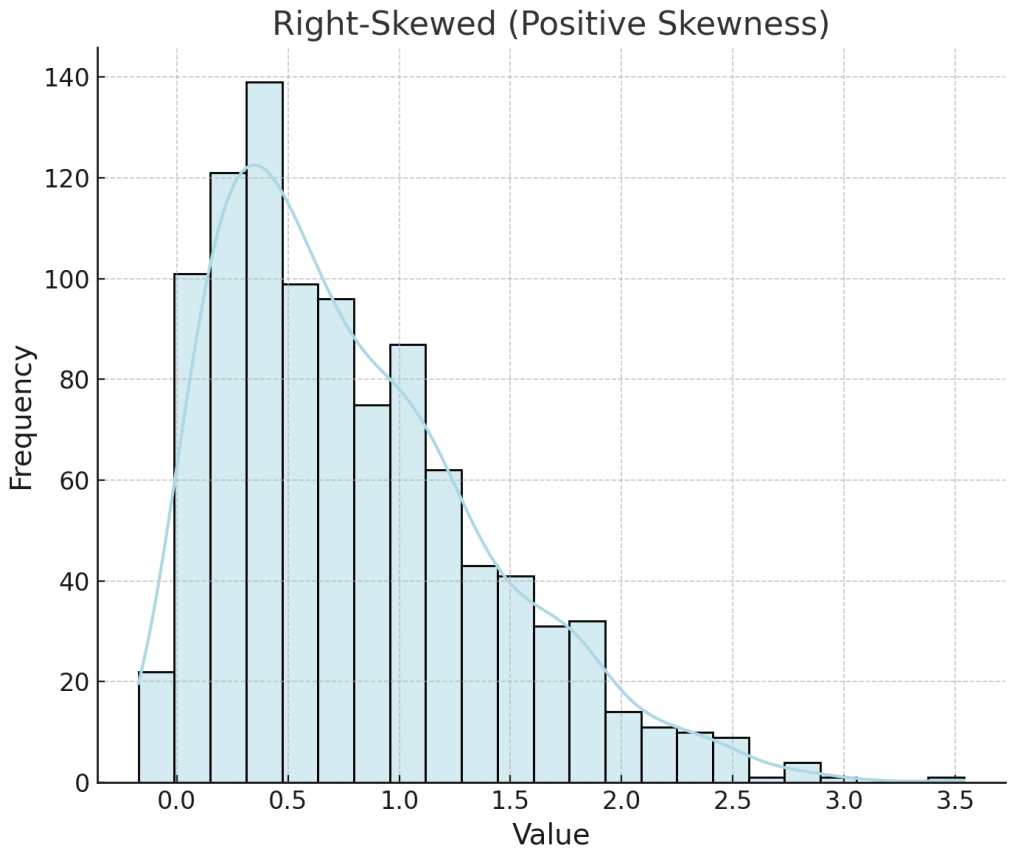
- Measures of Central Tendency
  - **Mean:** The average value of the data.
  - **Median:** The middle value when data is ordered.
  - **Mode:** The most frequent value
- Measures of Data Spread
  - **Standard Deviation**: Shows how data deviates from the mean.
  - **Variance**: The square of standard deviation, representing data spread.
- Indicators of Distribution Shape
  - **Skewness**: Indicates the asymmetry of the data distribution.
  - **Kurtosis**: Measures the tail heaviness of the distribution.

Zero skewness

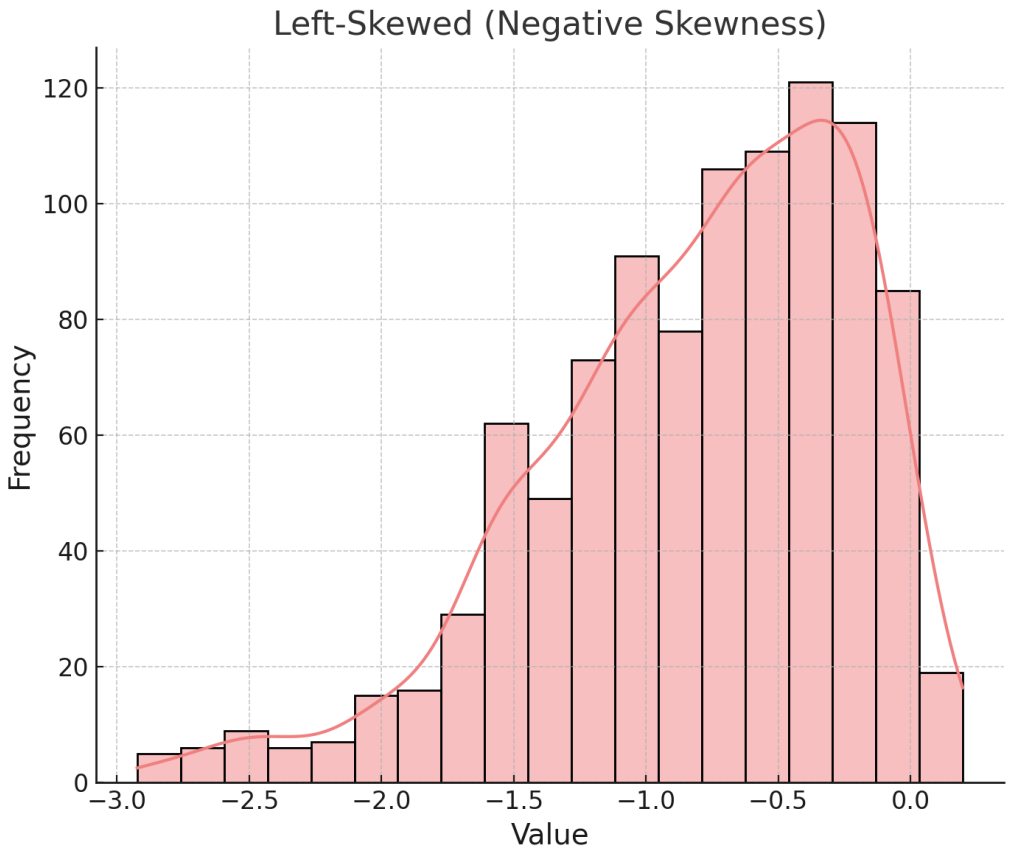


Mean = Median = Mode; (ideally)

Positive and Negative skewness



Mode < Median < Mean;
Long right tail



Mean < Median < Mode;
Long left tail

Kurtosis and its Types

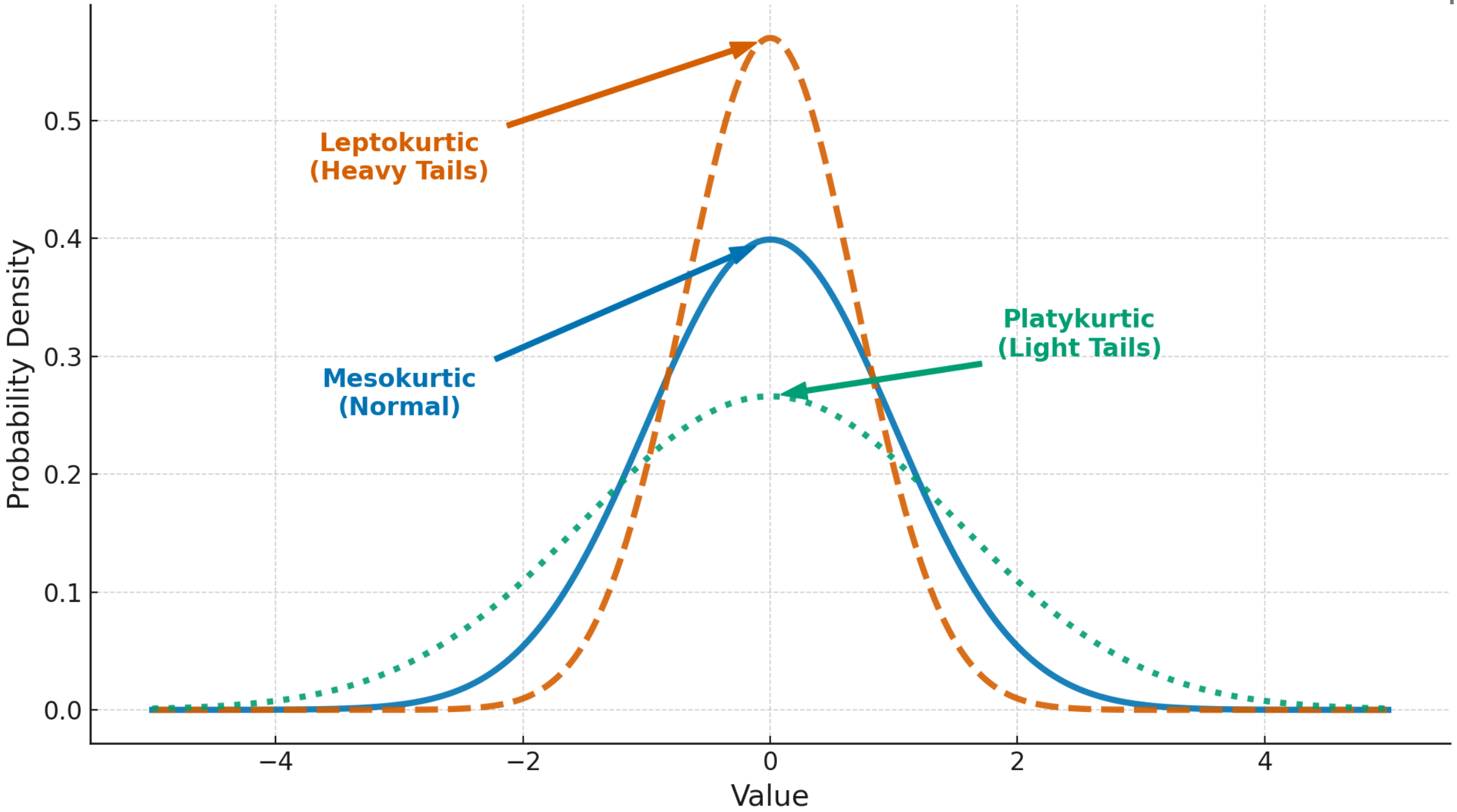
- **kurtosis** (from Greek: κυρτός, *kyrtos* or *kurtos*, meaning "curved, arching")

Distributions have three types of kurtosis:

- **Mesokurtic distribution** (kurtosis = 3, excess kurtosis = 0): perfect normal distribution or very close to it.
- **Leptokurtic distribution** (kurtosis > 3, excess kurtosis > 0): sharp peak, heavy tails*
- **Platykurtic distribution** (kurtosis < 3, excess kurtosis < 0): flat peak, light tails *

***Heavy tails** refer to probability distributions where extreme values (far from the mean) have a higher likelihood compared to normal distributions. Heavy tails distributions(e.g., **Pareto**, **Cauchy**) have more frequent and larger outliers.
Light tails distributions (e.g., **Uniform** or **Normal)** have fewer outliers and less probability mass in the tails.

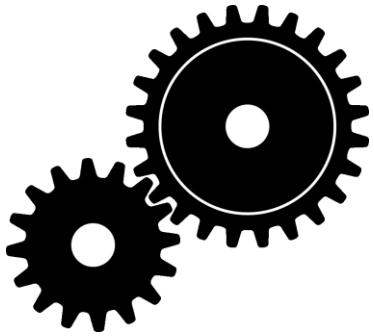
Illustration of Kurtosis Types



How to calculate statistics?

pandas NumPy SciPy

Libs



Custom Functions

Central Tendency: Mean, Median, Mode

```python
import pandas as pd

# Create a sample DataFrame
data = pd.Series([1, 2, 2, 3, 4, 5, 5, 6])

mean_pandas = data.mean()
median_pandas = data.median()
# mode returns a series, so take the first
mode_pandas = data.mode()[0]

print(f"Mean: {mean_pandas}")
print(f"Median: {median_pandas}")
print(f"Mode: {mode_pandas}")
```

```python
from collections import Counter

def calculate_mean(data):
    return sum(data) / len(data)

def calculate_median(data):
    sorted_data = sorted(data)
    n = len(sorted_data)
    mid = n // 2
    if n % 2 == 0:
        return (sorted_data[mid - 1] + sorted_data[mid]) / 2
    else:
        return sorted_data[mid]

def calculate_mode(data):
    counter = Counter(data)
    return counter.most_common(1)[0][0]

data = [1, 2, 2, 3, 4, 5, 5, 6]
mean_ = calculate_mean(data)
median_ = calculate_median(data)
mode_ = calculate_mode(data)

print(f"Mean: {mean_}")
print(f"Median: {median_}")
print(f"Mode: {mode_}")
```

Data Spread: Standard Deviation, Variance

```python
std_pandas = data.std()
variance_pandas = data.var()

print(f"Standard Deviation: {std_pandas}")
print(f"Variance: {variance_pandas}")
```

```python
def calculate_variance(data):
    mean_value = calculate_mean(data)
    return sum((x - mean_value) ** 2 for x in data) / (len(data) - 1)

def calculate_std(data):
    return calculate_variance(data) ** 0.5

std_custom = calculate_std(data)
variance_custom = calculate_variance(data)

print(f"Standard Deviation: {std_custom}")
print(f"Variance: {variance_custom}")
```

Indicators of Distribution Shape: Skewness, Kurtosis

```python
skewness_pandas = data.skew()
kurtosis_pandas = data.kurt()

print(f"Skewness: {skewness_pandas}")
print(f"Kurtosis: {kurtosis_pandas}")
```

```python
import numpy as np

def calculate_kurtosis(data):
    n = len(data)
    mean_value = np.mean(data)
    numerator = np.sum((data - mean_value) ** 4)
    denominator = np.sum((data - mean_value) ** 2) ** 2
    kurtosis = (n * numerator) / denominator
    return kurtosis

# Example data
data = [1, 2, 2, 3, 4, 5, 5, 6]

# Pearson's kurtosis
kurtosis_pearson = calculate_kurtosis(data)
print(f"Pearson's Kurtosis: {kurtosis_pearson}")

# Fisher's kurtosis (adjusted)
kurtosis_fisher = kurtosis_pearson - 3
print(f"Fisher's Kurtosis: {kurtosis_fisher}")
```

```
Pearson's Kurtosis: 1.628099173553719
Fisher's Kurtosis: -1.371900826446281
```

== Biased version ==

Pay attention to calculation!

bias=True

Bias correction might change the result

```python
import pandas as pd
import numpy as np
from scipy.stats import kurtosis

# Sample data
data = pd.Series([1, 2, 2, 3, 4, 5, 5, 6])

# Kurtosis using Pandas
kurtosis_pandas = data.kurt()
print(f"Kurtosis (Pandas): {kurtosis_pandas}")

# Kurtosis using SciPy (NumPy)
kurtosis_scipy = kurtosis(data, fisher=True, bias=True)
print(f"Kurtosis (SciPy - bias=True): {kurtosis_scipy}")

# Kurtosis using SciPy without bias correction
kurtosis_scipy_unbiased = kurtosis(data, fisher=True, bias=False)
print(f"Kurtosis (SciPy - bias=False): {kurtosis_scipy_unbiased}")

```

```
Kurtosis (Pandas): -1.4809917355371907
Kurtosis (SciPy - bias=True): -1.371900826446281
Kurtosis (SciPy - bias=False): -1.4809917355371898
```

Pay attention to calculation!

fisher=True

Fisher's definition subtracts **3** from the raw kurtosis value, so that the kurtosis of a normal distribution is **0**.

```python
from scipy.stats import kurtosis

# Sample data
data = [1, 2, 2, 3, 4, 5, 5, 6]

# Fisher's definition (default, kurtosis of normal distribution = 0)
kurtosis_fisher = kurtosis(data, fisher=True)
print(f"Kurtosis (Fisher's Definition): {kurtosis_fisher}")

# Pearson's definition (kurtosis of normal distribution = 3)
kurtosis_pearson = kurtosis(data, fisher=False)
print(f"Kurtosis (Pearson's Definition): {kurtosis_pearson}")

```

```
Kurtosis (Fisher's Definition): -1.371900826446281
Kurtosis (Pearson's Definition): 1.628099173553719
```

Key Statistical Measures in EDA (categorical data)

- **Frequency Count**:
  - The number of occurrences of each category.
  - Example: How many passengers belonged to each class on the Titanic (First, Second, Third Class)?
- **Proportion or Percentage**:
  - The relative frequency of each category as a proportion or percentage of the total.
  - This is useful for understanding the prevalence of each category.
- **Mode (or Top)**:
  - The most frequent category in the dataset.
  - For categorical data, the mode is often a key measure of central tendency.
- **Unique Count**:
  - The number of distinct or unique categories.
  - Example: The number of different destinations for airline passengers.
- **Missing Values**:
  - The count of missing or null values in the categorical column.
  - It’s important to track missing values before performing further analysis.

Live Coding & Demos #1

UNIVARIATE ANALYSIS

Exploring Single Variable Insights

Univariate analysis involves analyzing **one variable** at a time.

It aims to understand the **distribution, central tendency**, and **spread** of the variable.

Key Objectives:

- **Identify patterns**: Detect trends, anomalies, and outliers in the data.
- **Measure distribution**: Understand whether data is **normal**, **skewed**, or **bimodal**.
- **Aid in data cleaning**: Helps spot missing values or inconsistencies that need correction.

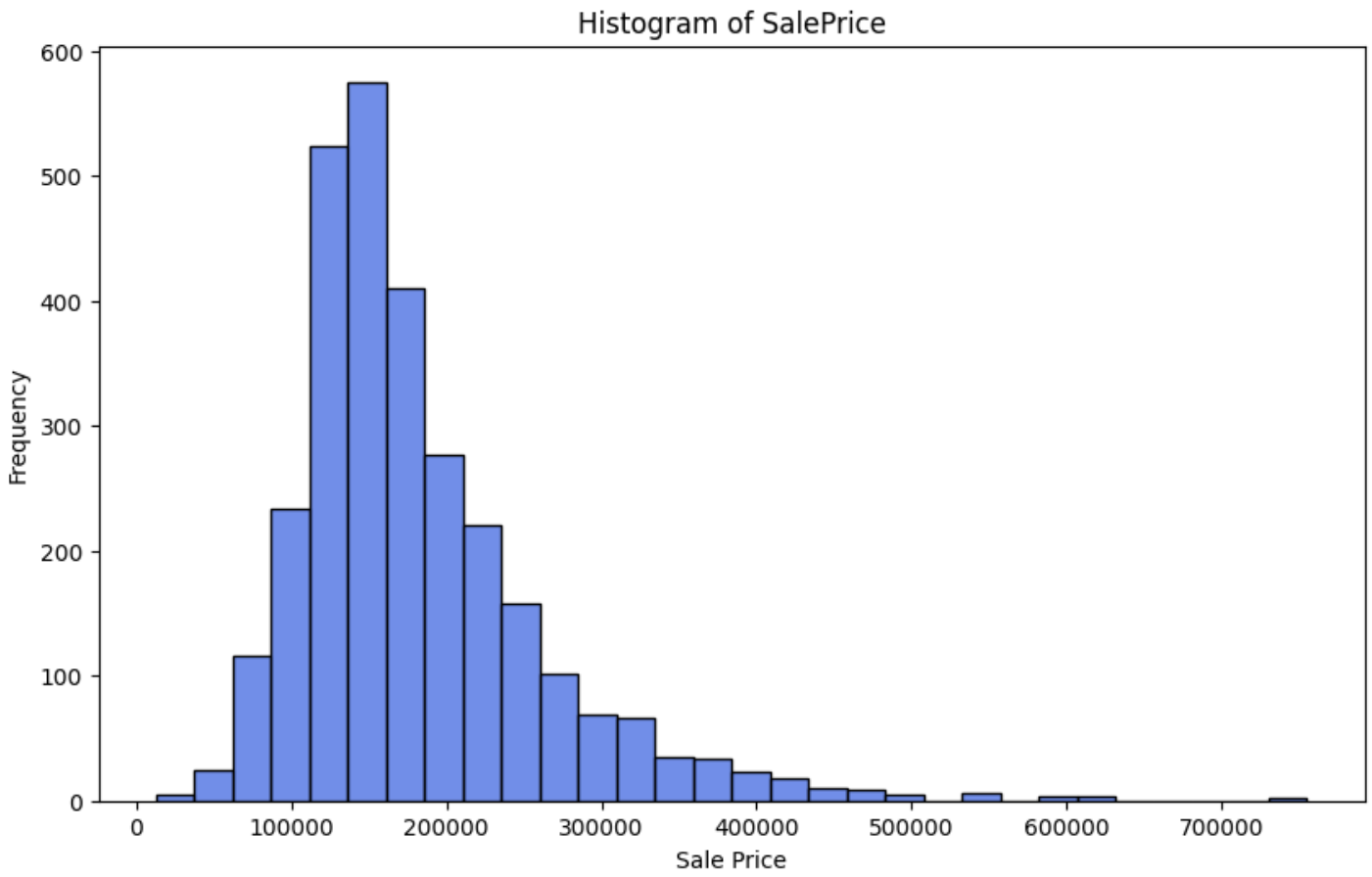
Univariate Analysis **Visualization Techniques**

- **Histogram**:
  - Visualizes the **frequency** of data values.
  - Best for identifying the **shape** of the data (e.g., normal, skewed).
- **KDE (Kernel Density Estimation)**:
  - A **non-parametric** method to estimate the probability density from the data.
  - Great for **visualizing the data's shape** without assuming any specific distribution.
- **ECDF (Empirical Cumulative Distribution Function)**:
  - Plots the **cumulative probability** directly from observed data.
  - Shows how data points **accumulate**, helpful for visualizing percentiles.
- **Boxplot**:
  - Displays the **five-number summary** (min, Q1, median, Q3, max) and highlights **outliers**.
  - Best for understanding **central tendency** and **spread**.
- **Violin Plot**:
  - Combines a **boxplot** with a **density estimate** to show the distribution's shape.
  - Visualizes **data density** and spread effectively.

Histogram

What is it?
A graphical representation that shows the **frequency distribution** of data points in specified intervals (bins).

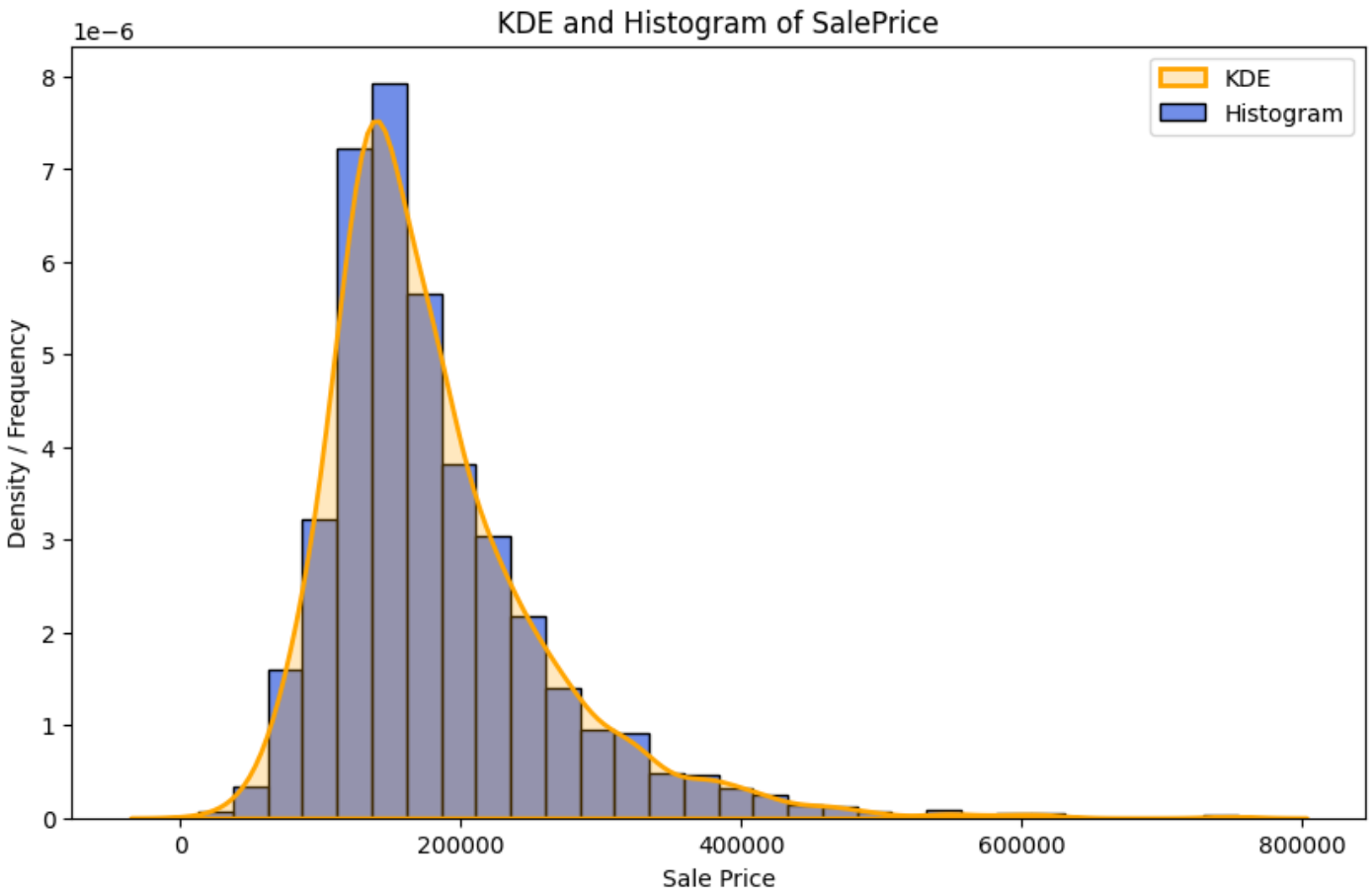
Purpose:
Helps visualize the shape of the data distribution (e.g., normal, skewed, or bimodal). Look for patterns such as **bell-shaped**, **skewness**, or **multiple peaks**.



KDE (Kernel Density Estimation)

What is it?
A **non-parametric** method for estimating the probability density function (PDF) of data.

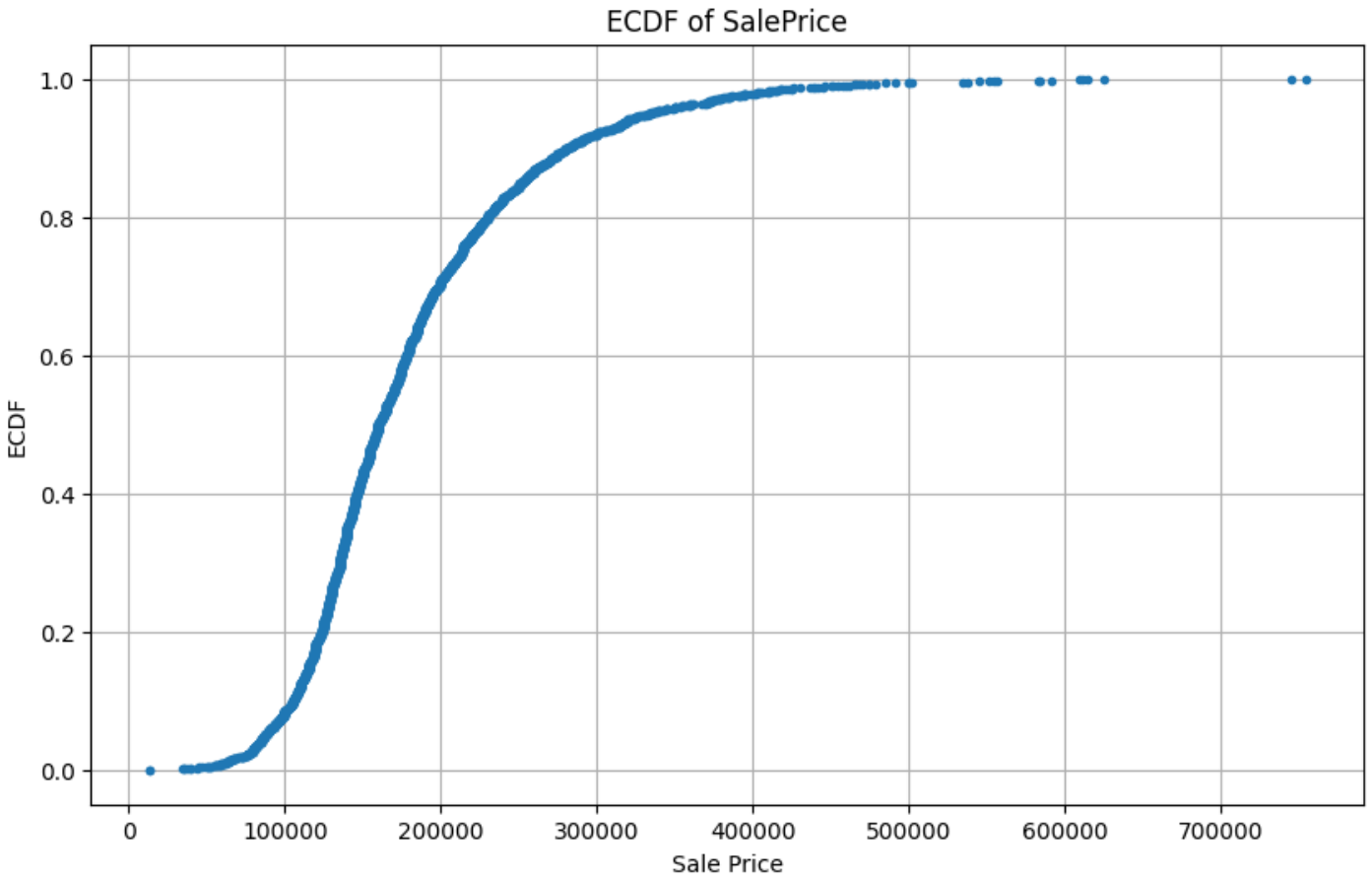
Purpose:
Provides a **smoothed curve** that estimates the underlying data distribution without assuming any specific distribution.
Ideal when you don't know the exact distribution of the data but want to visualize its shape.



ECDF (Empirical Cumulative Distribution Function)

What is it?
A function that shows the **cumulative probability** for each data point.

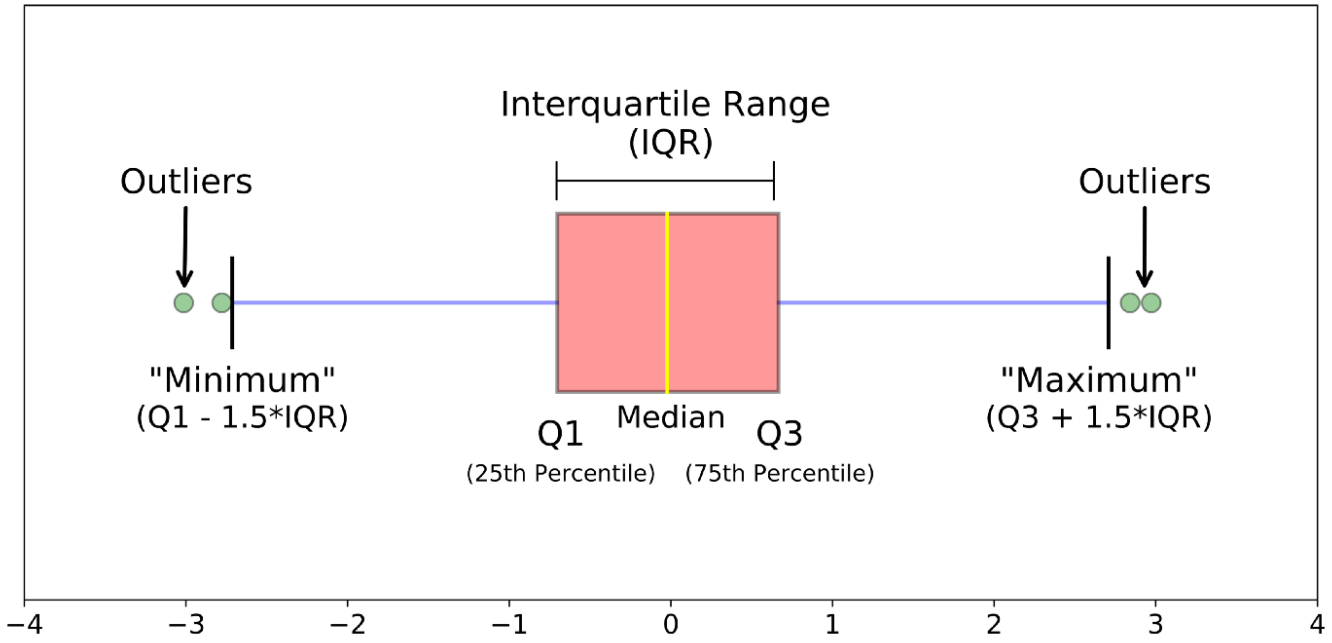
Purpose:
Visualizes how the **data accumulates**, useful for finding percentiles or understanding the rank of values.



Boxplot

What is it?
A plot that displays the **five-number summary**: minimum, Q1, median, Q3, maximum, and highlights **outliers**.

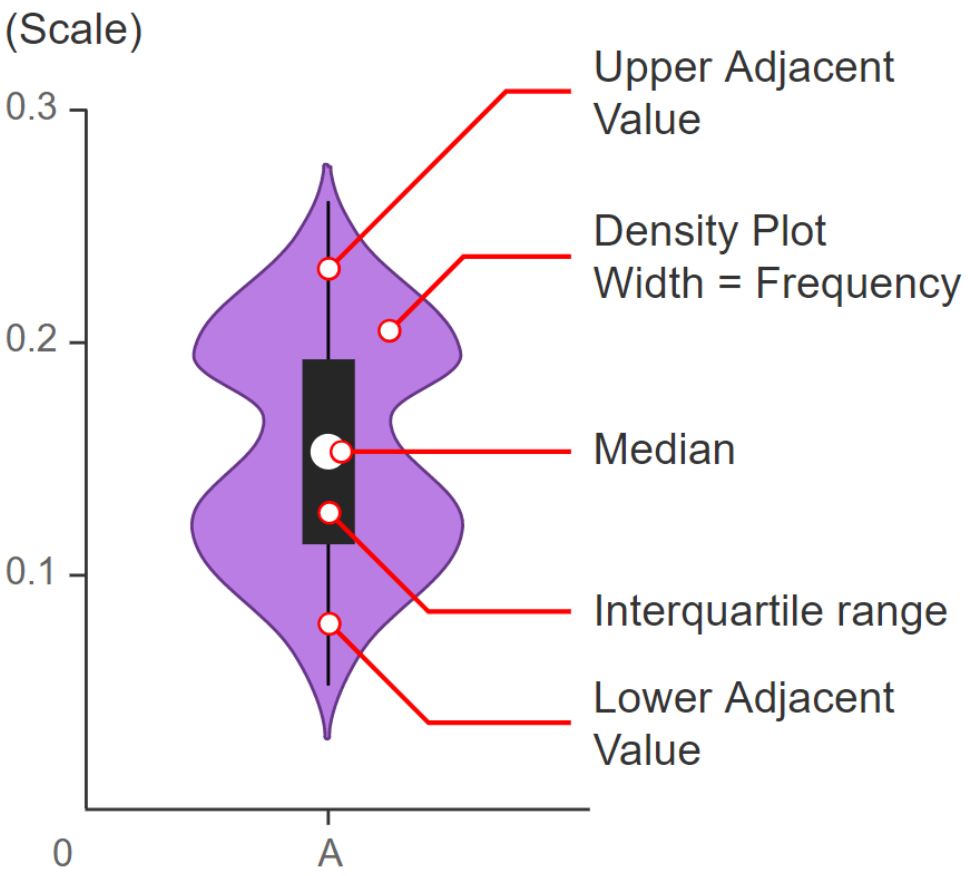
Purpose:
Helps visualize **central tendency**, **spread**, and potential **outliers**.
Boxplots help identify **outliers** and display **data spread** through the **interquartile range (IQR)**.



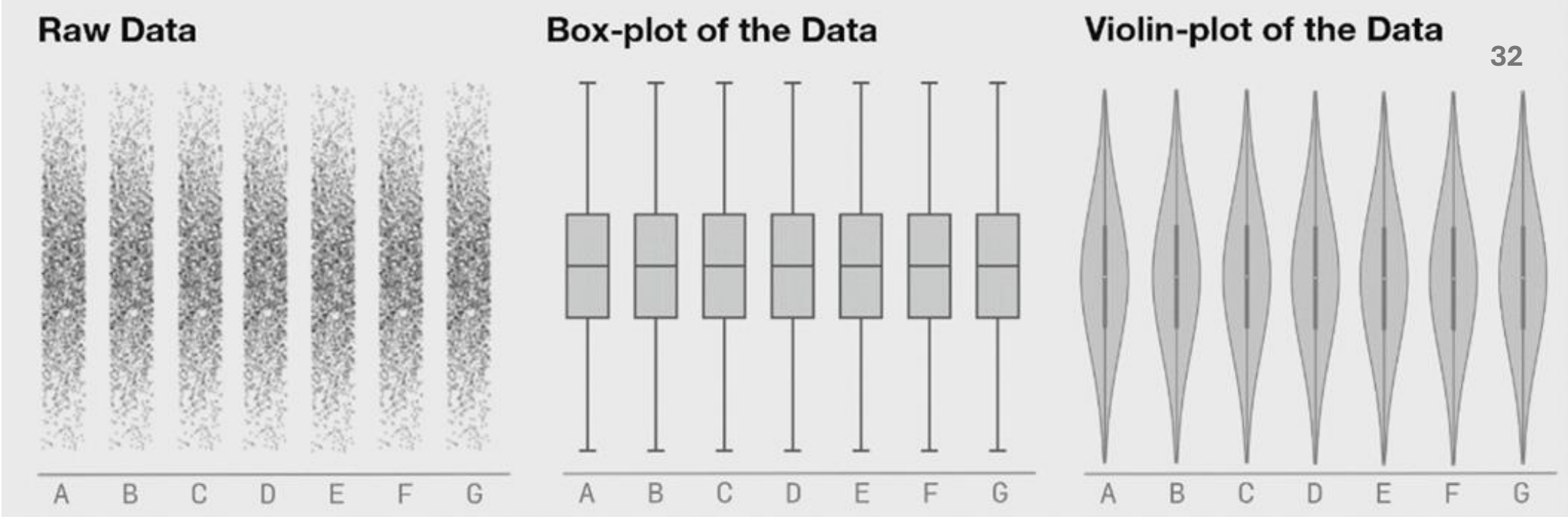
Violin plot

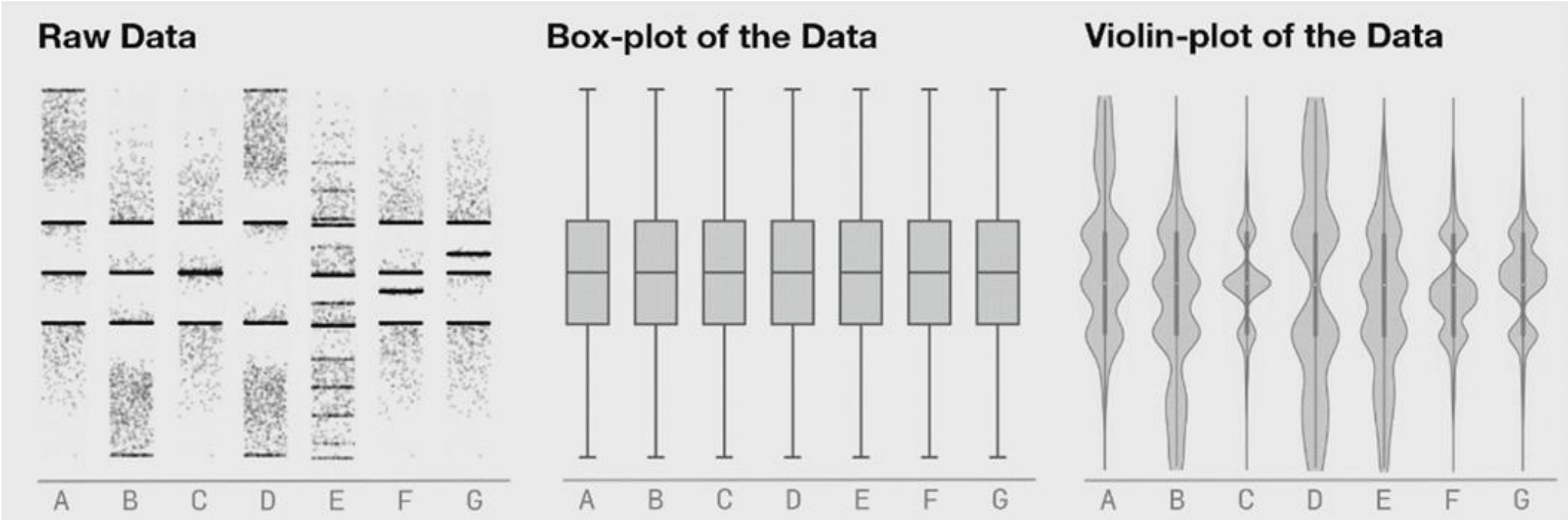
What is it?
Combines a **boxplot** with a **KDE** to visualize the distribution's shape.

Purpose:
Offers a more detailed view of the **data distribution density** compared to a boxplot alone.
Use it for **visualizing density** and spread in a single view.



Box plot & Violin plot





Live Coding & Demos #2

BIVARIATE ANALYSIS

Exploring Two Variables Insights

Bivariate analysis involves analyzing the **relationship between two variables**.

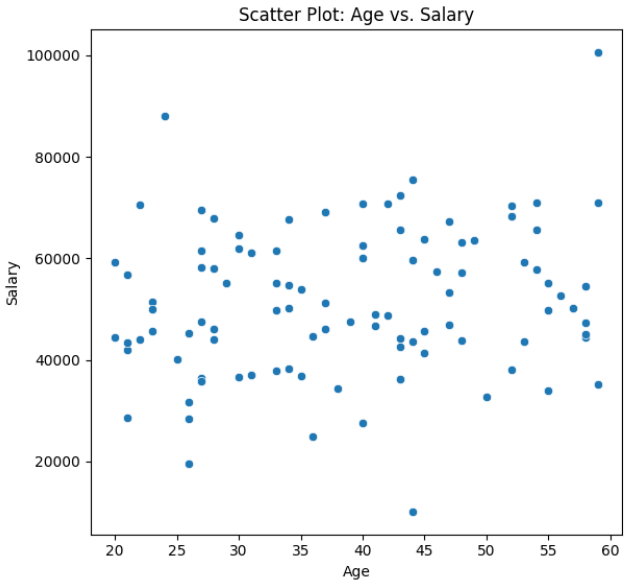
It aims to understand how two variables interact, whether one influences the other, and to detect patterns or associations.

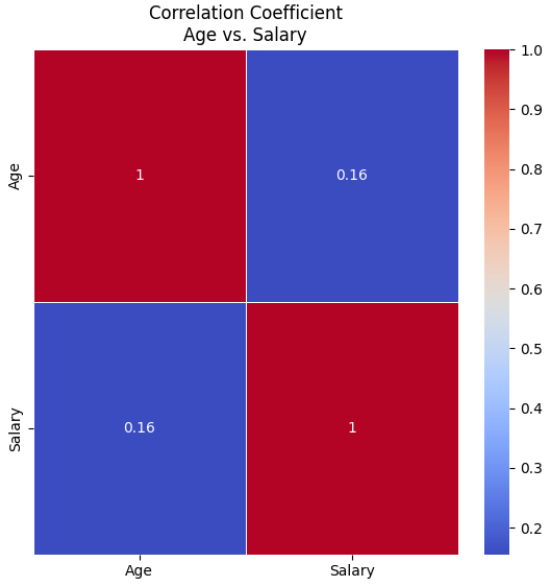
Key Objectives:

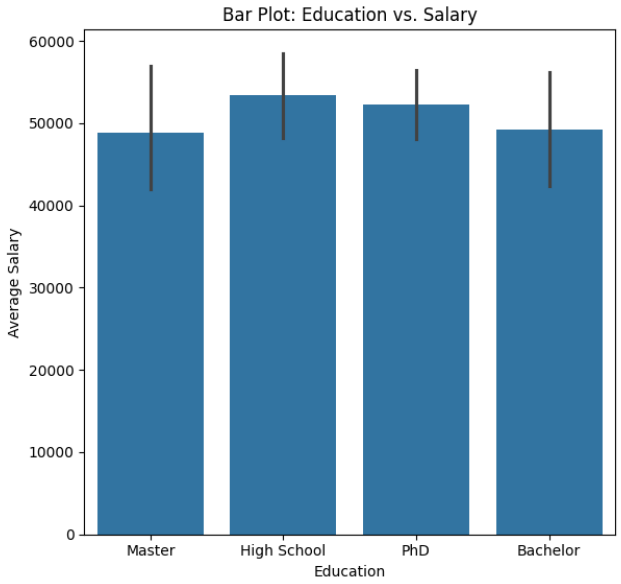
- identify **correlations**, **dependencies**, and **trends** between two variables.
- Provides insights into how **one variable impacts** another, which is useful for predictive models.
- **Essential** for regression, classification, and other predictive modeling tasks.

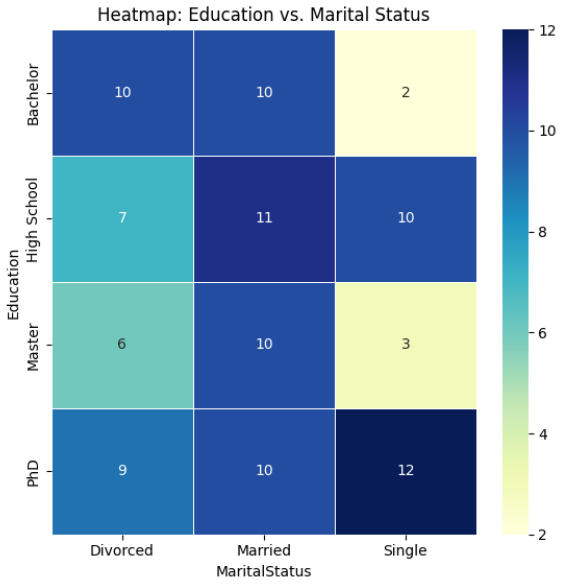
Bivariate Analysis **Visualization Techniques**

- **Quantitative Data:** Scatter plots, correlation coefficients, and regression techniques.
- **Categorical Data:** Bar charts, heatmaps, or cross-tabulation.
- **Mixed Data Types:** Techniques vary depending on the variable types.





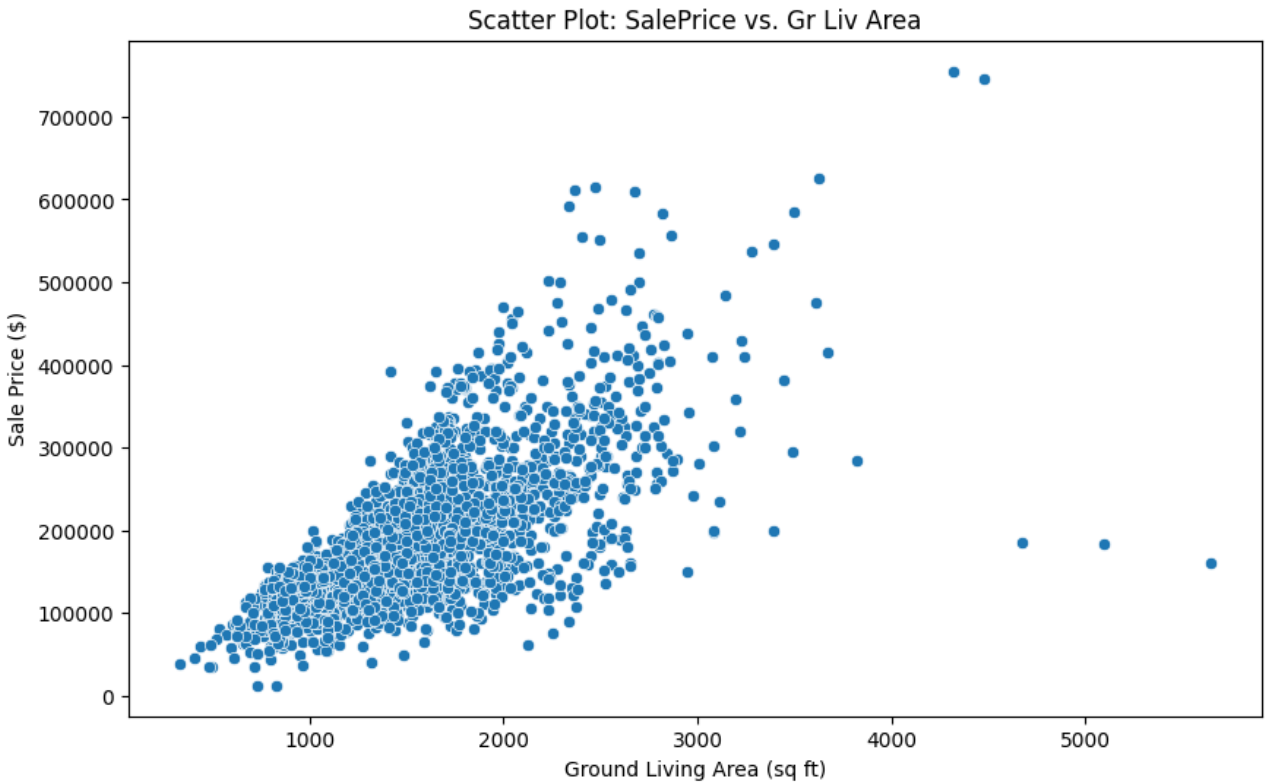




Scatter plot

What is it?
A graph that plots points to show the relationship between two **numerical variables**. Each axis represents one variable, and each point represents a data point.

Purpose:
Use it to identify if there is a **linear relationship** (positive or negative) between two variables. To detect **clusters** or **outliers** in the data.



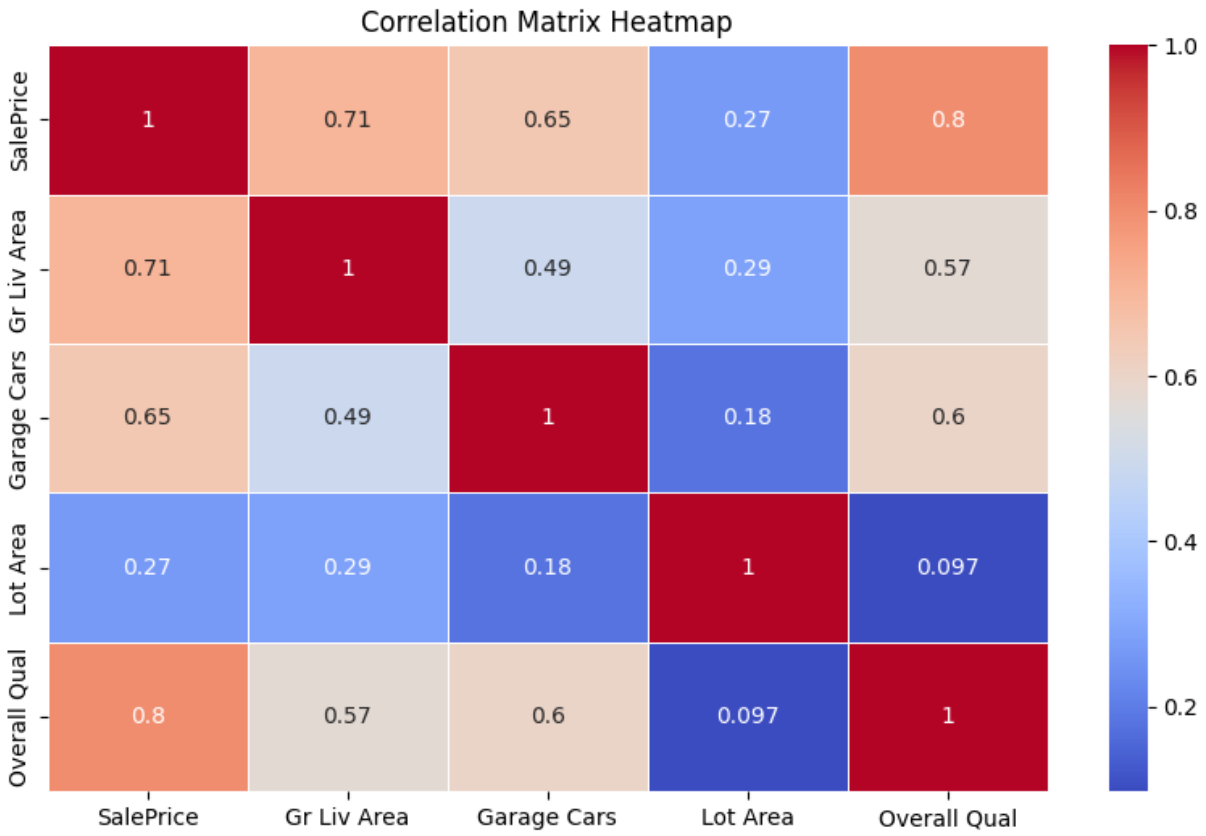
How to Interpret:
- **Positive Correlation**: Points slope upwards.
- **Negative Correlation**: Points slope downwards.
- **No Correlation**: Points are scattered randomly.

Correlation Matrix (Heatmap)

What is it?
A table showing the **correlation coefficients** between multiple variables. The coefficient ranges from -1 to 1, indicating the strength and direction of the relationship between variables.

Purpose:
Use it to quickly assess which pairs of variables are **correlated** and how strongly. Useful when working with **many variables** to detect patterns in one view.

How to Interpret:
1: Perfect positive correlation.
-1: Perfect negative correlation.
0: No correlation.



Cross-Tabulation (Contingency Table)

What is it?
Statistical tool used to analyze the relationship between **two categorical variables** by displaying the frequencies (or counts) of each combination of categories.

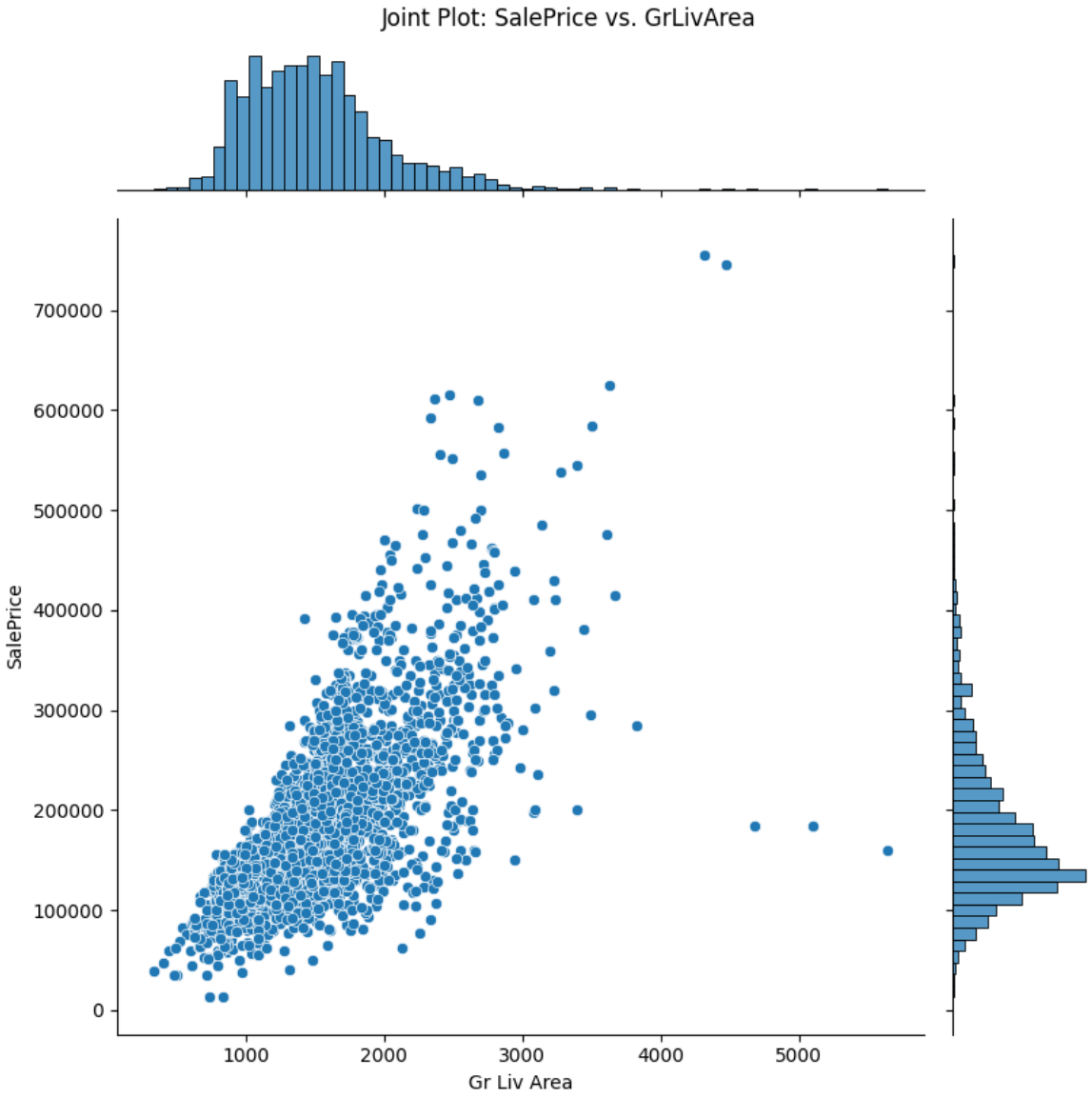
Purpose:
It helps to identify **patterns** and **interactions** between two categorical variables and is a starting point for further statistical tests.
Visualizes how frequently two categorical variables occur together. Identifies **dependencies** or **associations** between categorical variables.

High counts in specific cells indicate a stronger **association** between those categories.

Joint Plot

What is it?
A plot that combines a scatter plot with **histograms** or **KDE plots** for each variable along the axes..

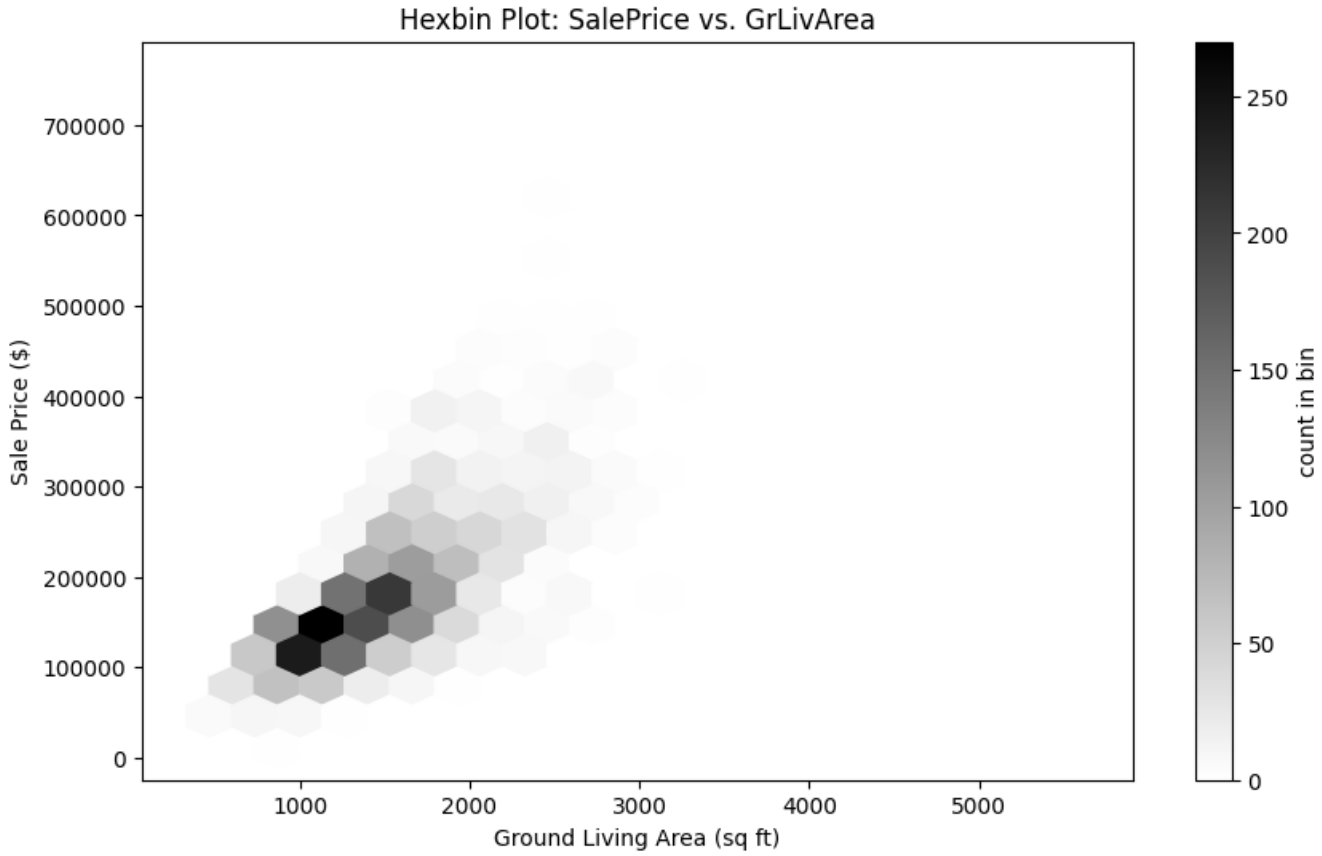
Purpose:
Provides both the **relationship** between two variables and their **individual distributions**. Use it to see how two variables are related while also understanding their **distribution**. To visualize potential **correlations** and **marginal distributions** in one view



Hexbin Plot

What is it?
A plot similar to a scatter plot but uses **hexagonal bins** to group data points.

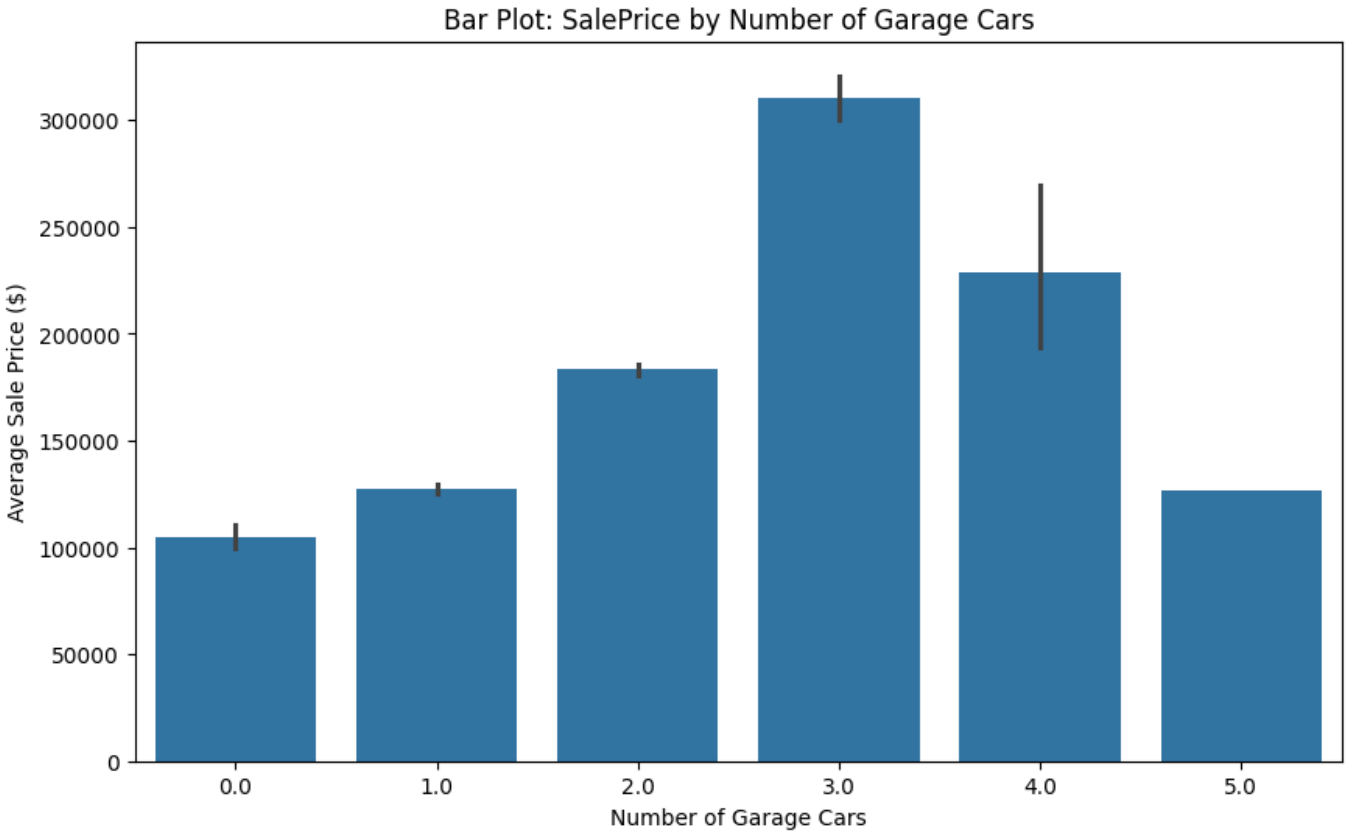
Purpose:
Helps visualize the **density** of data points, especially in large datasets where overplotting might occur in a scatter plot. Use it when working with **large datasets** and the scatter plot becomes too cluttered. Ideal for visualizing **density patterns** between two variables.



Bar Plot

What is it?
A plot used to show the **relationship between a categorical variable and a numerical variable**.

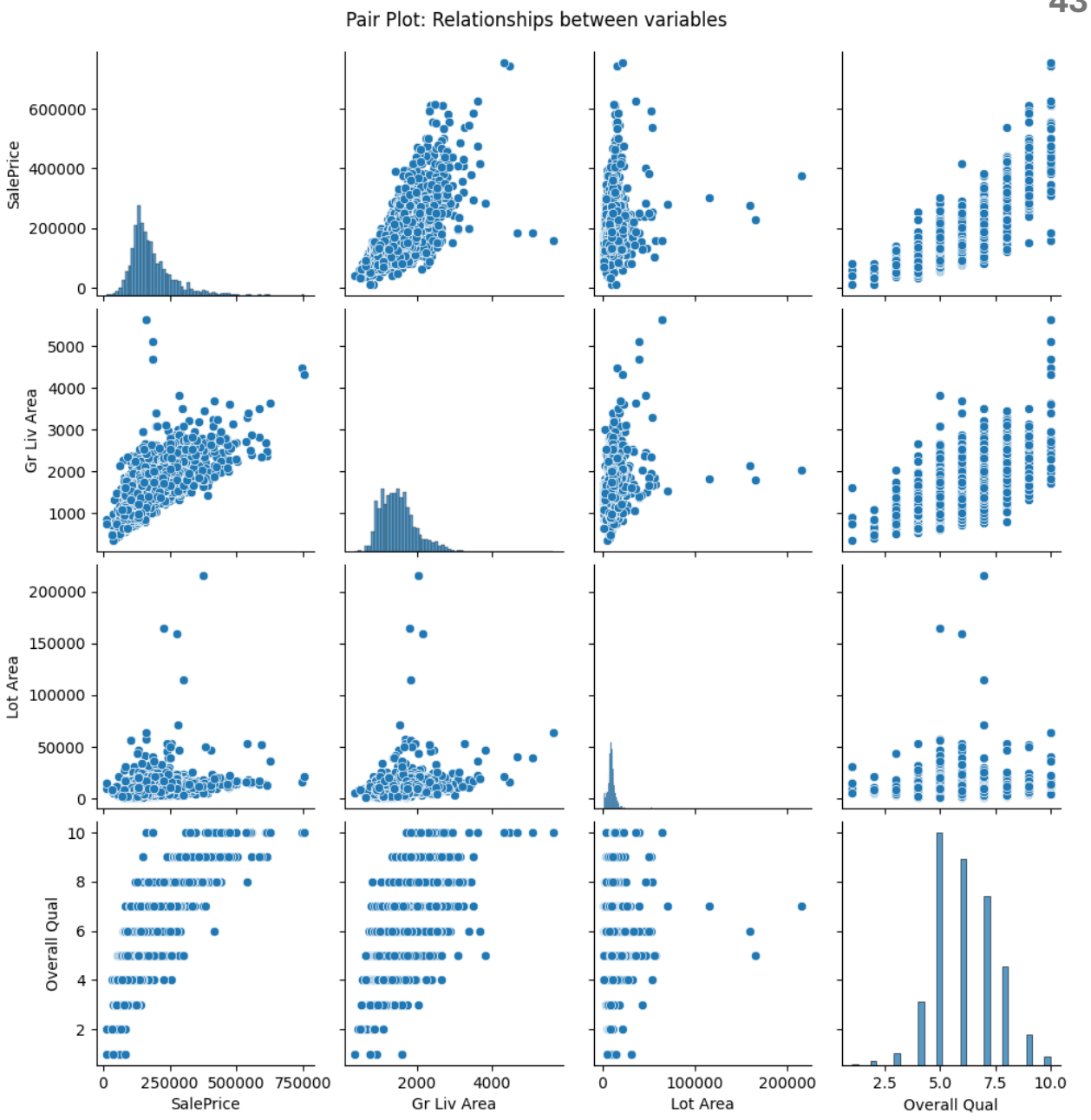
Purpose:
Helps compare the mean, sum, or count of the numerical variable across different categories. Use it to compare **groups** or **categories** against a numerical variable. Ideal for categorical data such as **gender vs. salary** or **education vs. income**.



Pair Plot

What is it?
A grid of scatter plots that shows the relationships between **all pairs of numerical variables** in a dataset.

Purpose:
To explore **pairwise relationships** in a dataset and identify potential **correlations** or **patterns**. Useful when working with **multiple numerical variables** to detect **correlations** across all combinations. Useful for **exploratory analysis**.



Live Coding & Demos #3

MULTIVARIATE ANALYSIS

Multivariate Analysis – Expanding Our View

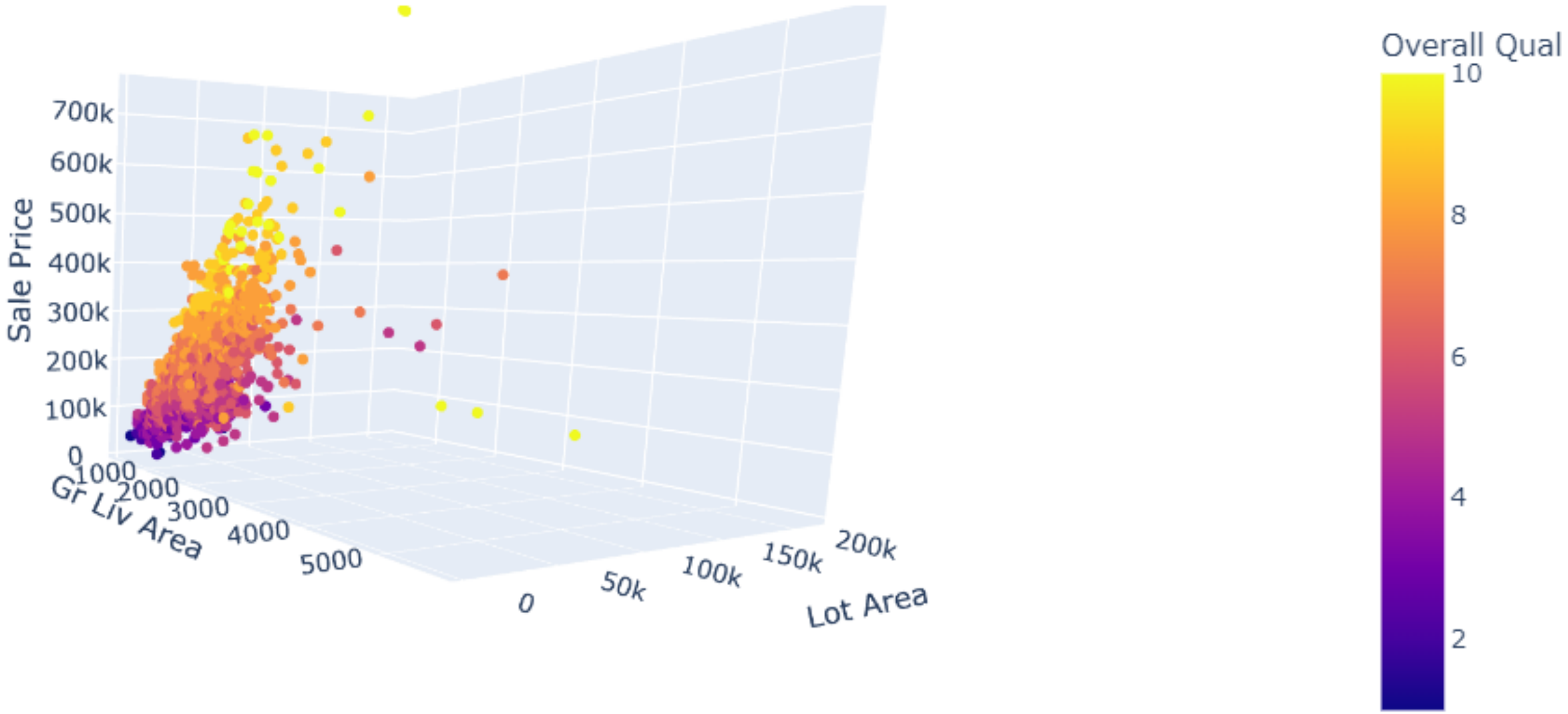
Moving beyond pairs of features:

- We've explored **single features** and **pairs of variables**.
- Multivariate analysis allows us to study **many variables at once**, revealing complex patterns and relationships.

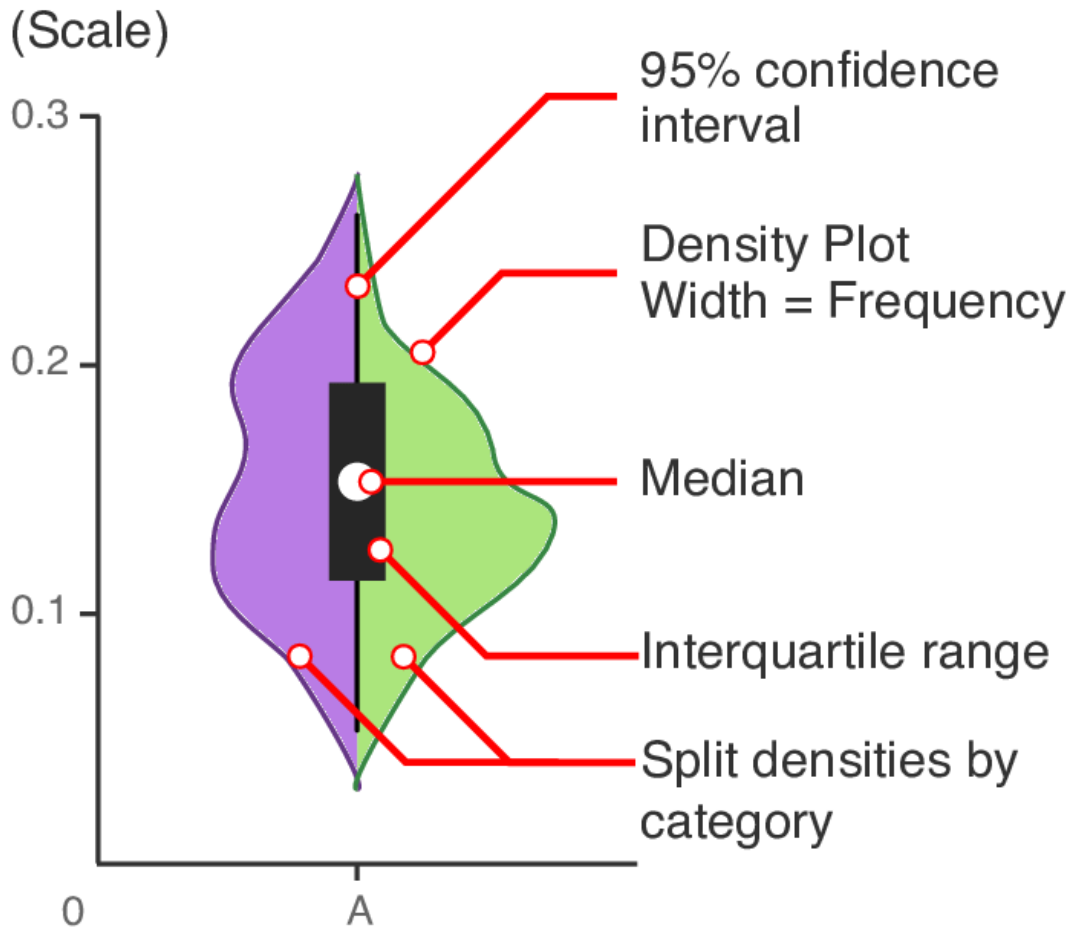
Why This Matters:

- By analyzing multiple features together, we can uncover **deeper insights** and **interactions** that are not visible in simpler analyses.
- This is key for building more **accurate models** and understanding how different variables influence each other.

3D Scatter Plot



Violin Plot for multivariate case



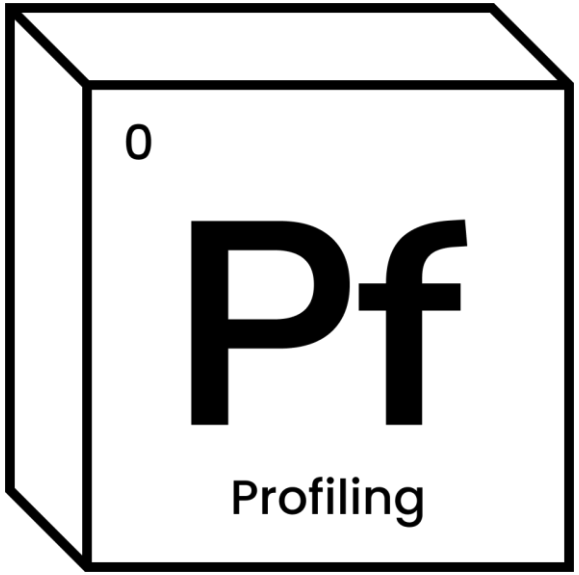
Live Coding & Demos #4

SweetVIZ

Automation in EDA

D-TALE

1000465	date	security_id	int_val	Col0	Col1	Col2	Col3	Col4	Col5	Col6	Col7	Col8	Col9	Col10	Col11
0	2018-09-13	100000	13700556066	1.65	1.68	-0.09	0.68	2.04	1.40	-0.74	-0.15	-0.40	-0.07	0.16	0.90
1	2018-09-13	100001	32370834703	0.81	-0.68	-2.29	-0.58	1.05	0.83	-0.01	0.40	-0.63	-0.34	-1.62	1.99
2	2018-09	2	74359132292	0.27	1.39	[illegible]	[illegible]	-1.1	27	-1.4	0.28	0.04	[illegible]	[illegible]	0.61
3	2018-09-13	[illegible]	3555690688	0.24	[illegible]	[illegible]	-0.31	[illegible]	1.95	[illegible]	0.95	1.2	[illegible]	-1.74	-0.48
4	2018-09-13	100004	105740597	-0.64	-1.38	0	1.07	[illegible]	0.44	[illegible]	-1.81	[illegible]	[illegible]	2.02	1.05
5	2018-09-13	100005	[illegible]	1.00	0.7	1.73	[illegible]	[illegible]	[illegible]	16	-0.89	[illegible]	1.18	0.41	1.03
6	2018-09-	[illegible]	30414943406	0.09	[illegible]	0.9	[illegible]	[illegible]	0	0.39	-1.16	[illegible]	[illegible]	1.76	1.11
7	[illegible]	100007	94115021503	0.07	[illegible]	0.8	0.60	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.16	-1.00	-1.23
8	2018-09-	100008	7207483836	-0.03	-0.68	-0.06	-0.58	[illegible]	[illegible]	-1.13	0.7	0.09	-1.13	0.07	-0.07
9	2018-09-13	100009	24054692457	0.78	0.28	0.66	0.88	0.7	-0.57	-0.51	-0.29	-0.71	0.48	-1.07	0.41
10	2018-09-13	100010	55729553089	-0.24	-0.04	-0.29	1.63	-2.35	-0.18	0.35	-1.19	1.96	-1.02	0.43	1.40
11	2018-09-13	100011	79679700690	-0.41	0.34	-1.21	0.17	-0.94	-0.35	-1.54	-0.74	-0.81	-1.05	0.57	-0.20
12	2018-09-13	100012	35404502206	0.26	0.93	0.09	-0.11	0.77	-0.41	0.25	-1.92	1.05	-1.74	-0.58	-0.36
13	2018-09-13	100013	54366289347	1.60	0.55	-1.83	-0.74	1.90	-1.42	0.52	-0.86	1.19	0.47	1.69	-0.17
14	2018-09-13	100014	29512587692	0.74	-1.22	-0.79	0.40	0.20	-0.61	-1.58	1.87	1.11	-0.48	0.21	0.12
15	2018-09-13	100015	59921981555	1.70	0.39	-0.51	-0.77	0.99	-0.69	0.04	-0.87	1.60	0.31	1.39	-0.09



Automating Exploratory Data Analysis (EDA)

Why Automate EDA?

- **Saves time and effort**: Automation tools expedite the EDA process, especially with large datasets.
- **Quick insights**: Provides immediate understanding of data structure, trends, and anomalies.
- **Consistency**: Ensures comprehensive and uniform analysis across datasets.

Key Tools for Automating EDA

- **pandas_profiling**: Generates interactive HTML reports from pandas DataFrames.
- **Sweetviz**: Detailed comparison reports, ideal for comparing datasets (e.g., train/test splits).
- **D-Tale**: Interactive data exploration with GUI for pandas DataFrames.

Features of EDA Automation Tools

Tool	Key Features	Visualization Capabilities
ydata-profiling (ex. pandas_profiling)	- Data types, missing values, quantile stats (min, max, quartiles) - Descriptive stats (mean, mode, skewness)	- Histograms - Correlation heatmaps (Pearson, Spearman, Kendall) - Missing values matrix
Sweetviz	- Focus on dataset comparison (e.g., train/test) - Target analysis - Frequent values	- Visual comparisons - Correlation heatmaps - Target variable analysis
D-Tale	- Interactive interface for pandas - Filtering, sorting, and exploring data - Analytics tools	- Histograms - Correlation heatmaps - Interactive data exploration

Automating EDA – Common Pitfalls

Why You Shouldn't Rely Entirely on Automated EDA:

- **Lack of Context**: Automated tools don't understand the specific **context** or **your goals**
- **Overlooking Nuances**: They may miss **subtle patterns**, correlations, or data quality issues that require human insight.
- **No Critical Thinking**: It can promote a **passive approach** to data exploration, leading to missed opportunities for deeper, hands-on analysis.

Use automated EDA tools as a **starting point**, but always combine them with **manual exploration** to uncover deeper insights and drive better decision-making.

Live Coding & Demos #5

Visualization Tools and Resources

Visualizations in EDA

Importance of visualization:
- Enhances **understanding** and **communication** of data insights.
- Crucial for **storytelling** and **decision-making**.
- Ensures **comprehensive** and **uniform analysis** across datasets.

Storytelling with Matplotlib:
- **Customization** for clarity and impact.
- Add **titles**, **labels**, **legends**, and **annotations**.
- Adjust **styles** and **color schemes** to improve aesthetics.

Visualization Resources:

Data Visualization Catalogue:
https://datavizcatalogue.com/index.html
- Useful resource for choosing the best **chart types**.
- Helps in selecting the most **appropriate visualization**.

COLOR SCHEMES FOR COLOR BLIND:
- https://davidmathlogic.com/colorblind/#%23D81B60-%231E88E5-%23FFC107-%23004D40
- https://www.datylon.com/blog/data-visualization-for-colorblind-readers
- https://seaborn.pydata.org/tutorial/color_palettes.html

Advanced Visualization Libraries:

Library	Key Features	Best Use Cases
Seaborn	- Built on Matplotlib - Simplifies heatmaps, time series plots	- Statistical graphics - Attractive and informative plots
Plotly	- Interactive, web-based visualizations - Supports 3D plotting	- Dashboards - Presentations requiring user interaction
Altair	- Declarative syntax - Quick and easy interactive plots	- Small to medium datasets - Interactive plots
Bokeh	- High-performance, interactive visualizations - Real-time data	- Large datasets - Real-time streaming and dashboards
ggplot (Python)	- Grammar-based approach to visualizations - Layered control	- Complex plots - Similar to ggplot2 in R

Live Coding & Demos #6

Common EDA Challenges

Addressing Key Challenges in EDA

- **Dealing with Missing Data:**
  - Missing data can skew results and affect model performance.
- **Managing Inconsistent Data:**
  - Inconsistent formats (e.g., date formats, units) cause errors in analysis.
- **Identifying Outliers:**
  - Outliers can distort analysis and affect the accuracy of models.
- **Handling High Dimensionality:**
  - Too many variables can lead to **overfitting** and make interpretation difficult.
- **Addressing Data Imbalance:**
  - Imbalanced data (e.g., one class dominating) can result in biased models.

Upcoming in the Data Preparation Lecture:

We will **dive deeper** into each of these challenges during the **Data Preparation** lecture.
Learn **practical techniques** for handling these challenges, ensuring **data quality** and **reliability** in your analyses.

Final Remarks and Recap

Recap: Exploratory Data Analysis

We reviewed **data inspection**, **univariate**, **bivariate**, and **multivariate analysis**. Covered various **visualization** techniques.

EDA helps in **understanding your data before modeling**.

It helps make **decisions** and prevents analytical mistakes.

EDA is an ongoing, **iterative process**

Practice with diverse datasets.

Use EDA to guide **data cleaning** and **feature selection**.

Source: https://xkcd.com/1838/

Useful Resources:

- CRISP-DM: https://www.datascience-pm.com/crisp-dm-2/
- Data Visualization Catalogue: https://datavizcatalogue.com/index.html
- From Data to Viz: https://www.data-to-viz.com/
- How to choose the right data visualization (by Atlassian): link
- UCLA Data Visualization: https://guides.library.ucla.edu/data-visualization
- Seaborn: https://seaborn.pydata.org/
- Plotly: https://plot.ly/
- ydata_profiling: https://github.com/ydataai/ydata-profiling
- Sweetwiz: https://github.com/fbdesignpro/sweetviz
- D-Tale: https://github.com/man-group/dtale
- **Kaggle**: A platform for practice with real datasets, providing kernels and notebooks that often include EDA. https://www.kaggle.com/

Thank you for your attention